UNIVERSITY OF CALIFORNIA

Los Angeles

Stochastic Discount Factors Implied by Machine Learning Forecasts of Equity Returns

A thesis submitted in partial satisfaction

of the requirements for the degree

Master of Applied Statistics and Data Science

by

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ABSTRACT OF THE THESIS

Stochastic Discount Factors Implied by Machine Learning Forecasts of Equity Returns

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This thesis examines whether different machine learning models, when used to forecast firm-level equity returns, imply materially different stochastic discount factors, and whether those differences translate into economically meaningful variation in firm-level risk exposures and pricing errors. The analysis compares linear and nonlinear models including ordinary least squares regression, least absolute shrinkage and selection operator (LASSO) regression, principal components regression, gradient boosted decision trees, feedforward neural networks, and long short-term memory (LSTM) networks under a unified empirical design.

Using monthly excess returns for S&P 500 constituent stocks from the Center of Research in Security Prices over the period 2000 to 2024, with predictors drawn from standard factor models and return-based anomalies, expected returns are first estimated at the firm level. These forecasts are then mapped into traded, time-varying stochastic discount factors via mean–variance efficient portfolio constructions derived from unconditional moment conditions. Model performance is evaluated using out-of-sample cross-sectional R^2 and average absolute pricing errors.

Across all specifications, the estimated stochastic discount factors are well behaved and

economically sensible. Nonlinear models generate more state-contingent and heavy-tailed pricing kernels than their linear counterparts, reflecting richer dynamics in state pricing. However, these differences do not translate into substantial improvements in firm-level pricing accuracy, with average pricing errors remaining similar across model classes. This evidence suggests that model choice primarily affects the dynamics and distributional properties of the pricing kernel rather than overall cross-sectional explanatory power.

Overall, the findings indicate that while machine learning methods meaningfully alter the implied structure of the stochastic discount factor, gains from nonlinear modeling are limited in terms of average firm-level pricing performance. A key limitation is the reliance on mean–variance efficiency in stochastic discount factor construction, as a nontrivial subset of firms exhibit idiosyncratic behavior that is difficult to price accurately within this framework. These results highlight the distinction between modeling state-dependent pricing dynamics and improving cross-sectional pricing accuracy, and point to directions for future work that relax portfolio efficiency assumptions or incorporate alternative pricing restrictions.

The thesis of Nils Berzins is approved.

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Contents

1 Introduction **1**

1.1 Motivation 1

1.2 Research Objective 2

1.3 Contributions 2

1.4 Organization of the Thesis 2

2 Literature Review **4**

2.1 Stochastic Discount Factors and No-Arbitrage Foundations 4

2.2 Test Assets, Anomalies, and the Role of Anomaly Portfolios 6

2.3 High-Dimensional SDF Estimation and Regularization 7

2.3.1 Linear factor models and the mean-variance efficient SDF 7

2.3.2 Regularization and model selection in asset pricing 7

2.3.3 Implications for the benchmark design 7

2.4 Machine Learning in Empirical Asset Pricing 8

2.4.1 Why ML methods are attractive in asset pricing 8

2.4.2 Prediction versus pricing objectives 8

2.5 Deep Learning Approaches to Learning the SDF 9

2.5.1 Learning pricing objects from return forecasts 9

2.5.2 Learning an SDF from large information sets 9

2.5.3 Imposing economic structure or no-arbitrage discipline 10

2.5.4 Evaluation practices in the literature 10

2.5.5 Where anomaly portfolios fit in this literature 11
2.6 Summary and Positioning of This Thesis 12
2.6.1 What this thesis does 12
2.6.2 How it differs from prior work 12

3 Data **13**
3.1 Data Sources and Construction 13
3.2 Sample Construction and Filtering 14
3.2.1 Sample period and universe 14
3.2.2 Rolling train–validation–test windows 14
3.2.3 Data cleaning and alignment 15
3.3 Descriptive Statistics 15

4 Methodology and Empirical Results **17**
4.1 Overview and Empirical Roadmap 17
4.2 Return Forecasting Models and Out-of-Sample Design 18
4.3 From Expected Returns to the Stochastic Discount Factor 19
4.4 Properties of Estimated Stochastic Discount Factors 19
4.5 Firm-Level Risk Exposures and Pricing Performance 21

5 Discussion **24**
5.1 Purpose and Overview 24
5.2 Summary of Empirical Findings 25
5.2.1 Properties of the Estimated Stochastic Discount Factors 25
5.2.2 Firm-Level Risk Exposures 25
5.2.3 Pricing Errors and Cross-Sectional Fit 26
5.3 Economic Interpretation 26
5.3.1 Economic Meaning of the Stochastic Discount Factor 26

5.3.2 Why Model Complexity Affects SDF Dynamics More Than Pricing Accuracy . . . 27
5.4 Relation to the Literature . . . 28
5.4.1 Machine Learning, Return Prediction, and No-Arbitrage Pricing . . . 28
5.4.2 Linear Benchmarks and Factor Models . . . 28
5.5 Robustness, Limitations, and Identification . . . 29
5.5.1 Sample Construction and Asset Coverage . . . 29
5.5.2 Model Dependence and Regularization . . . 29
5.5.3 Identification Limits . . . 29
5.6 Implications for Asset Pricing and Machine Learning . . . 30
5.6.1 Implications for Empirical Asset Pricing and Machine Learning . . . 30
5.7 Directions for Future Research . . . 30
5.8 Conclusion . . . 31

Supporting Materials **32**

A Model Notation and Assumptions . . . 32
A.1 Linear Prediction Models . . . 32
A.2 Nonlinear Prediction Models . . . 33
B The Stochastic Discount Factor . . . 34
B.1 What Is the Stochastic Discount Factor? . . . 34
B.2 How to Interpret and Contextualize the Stochastic Discount Factor . 34

Chapter 1

Introduction

1.1 Motivation

Asset pricing theory centers on the existence of a stochastic discount factor (SDF) that prices all assets in an economy. While classical linear factor models provide tractable representations of the SDF, empirical evidence suggests that linear specifications may struggle to capture nonlinearities and conditional dynamics present in financial return data. At the same time, recent advances in machine learning have introduced flexible modeling tools capable of approximating complex functional relationships, motivating renewed interest in their application to asset pricing problems.

Despite growing empirical adoption, the role of machine learning methods in SDF estimation remains an open question. In particular, it is unclear whether increased model flexibility leads to meaningful improvements in pricing performance once models are evaluated out of sample and under a common empirical design. This thesis contributes to this discussion by systematically comparing traditional linear approaches and modern machine learning models within a unified SDF estimation framework.

1.2 Research Objective

The primary objective of this thesis is to evaluate whether machine learning-based approaches to SDF estimation improve empirical asset pricing performance relative to standard linear benchmarks. Specifically, the analysis compares linear regression-based methods with non-linear machine learning models under identical data, training, and evaluation protocols.

Rather than focusing on predictive accuracy alone, the study emphasizes economically relevant criteria, including out-of-sample pricing errors and portfolio-level performance measures. By holding constant the information set and evaluation design, the analysis isolates the incremental contribution of model flexibility in SDF construction.

1.3 Contributions

This thesis makes three main contributions. First, it implements a unified empirical framework for estimating stochastic discount factors using both traditional econometric models and modern machine learning methods. Second, it provides a systematic comparison of these models using consistent out-of-sample evaluation metrics commonly employed in the asset pricing literature. Third, it offers evidence on the practical trade-offs associated with flexible SDF estimation, including potential gains in pricing performance and limitations related to stability, interpretability, and robustness.

Importantly, the focus of the analysis is comparative rather than prescriptive. The results are intended to inform the conditions under which increased model complexity may be warranted in empirical asset pricing applications.

1.4 Organization of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on stochastic discount factor estimation, linear factor models, and the use of machine learning

techniques in asset pricing. Chapter 3 describes the data sources, sample construction, and preprocessing steps used in the empirical analysis. Chapter 4 outlines the methodological framework, including model specifications, training procedures, and evaluation metrics as well as presenting the empirical results. Chapter 5 discusses the implications of the findings and concludes.

Chapter 2

Literature Review

2.1 Stochastic Discount Factors and No-Arbitrage Foundations

Stochastic discount factor (SDF). Asset pricing theory implies that in the absence of arbitrage, there exists a strictly positive stochastic discount factor (SDF), denoted m_{t+1}, such that all asset returns satisfy the fundamental pricing relation

$$\mathbb{E}_t[m_{t+1} R_{t+1,i}] = 1, \tag{2.1}$$

where $R_{t+1,i}$ denotes the gross return on asset i and $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ denotes the conditional expectation given the information set $\mathcal{F}_t$ available at time t. When returns are expressed in excess of the risk-free rate, this condition can equivalently be written as

$$\mathbb{E}_t\Big[m_{t+1} R^e_{t+1,i}\Big] = 0. \tag{2.2}$$

The SDF summarizes how states of the world are priced: assets earn risk premia because their payoffs covary with the SDF across states.

Many familiar asset pricing models can be interpreted as imposing structure on the SDF.

In linear factor models, the SDF is assumed to lie in the span of a small number of factor returns, while in consumption-based models it is linked to marginal utility growth. From this perspective, estimating an asset pricing model is equivalent to estimating a candidate SDF that prices a given set of assets.

No-arbitrage restrictions and empirical targets. The no-arbitrage restriction imposes moment conditions that discipline empirical asset pricing models. Given a candidate SDF, pricing errors arise when the empirical analog of the pricing equation fails to hold for a set of test assets. These pricing errors provide a natural metric for evaluating the economic validity of an estimated SDF.

In empirical applications, the SDF is typically evaluated relative to a finite collection of assets or portfolios. The choice of test assets therefore plays a critical role in determining how tightly the no-arbitrage restriction is enforced. A candidate SDF that prices a large and diverse set of assets well is subject to stronger economic discipline than one evaluated on a narrow subset.

Implications for empirical design. A key implication of the SDF framework is the distinction between pricing and prediction objectives. In efficient markets, realized returns are dominated by unforecastable shocks, and accurate prediction of realized returns is neither necessary nor sufficient for successful asset pricing. Instead, asset pricing focuses on explaining differences in expected returns through systematic exposure to the SDF.

This distinction motivates an evaluation framework centered on pricing-based criteria rather than predictive loss alone. In this thesis, candidate models are assessed based on their implied stochastic discount factors and evaluated using out-of-sample pricing errors and cross-sectional explanatory power. This approach aligns the empirical analysis directly with the no-arbitrage restrictions that underpin asset pricing theory.

2.2 Test Assets, Anomalies, and the Role of Anomaly Portfolios

Empirical asset pricing models are evaluated relative to a chosen set of test assets, and the informativeness of these assets plays a critical role in model assessment. Test assets that aggregate firm-level information into portfolios can reduce idiosyncratic noise and sharpen economically meaningful variation in expected returns. As a result, portfolio-level test assets are commonly used to evaluate the performance of candidate stochastic discount factors.

A large literature documents return patterns associated with firm characteristics, often referred to as anomalies. While early studies interpreted these patterns as evidence of mispricing, subsequent work has emphasized concerns related to data mining, instability, and replication. For example, Hou et al. (2020) show that the magnitude of many reported anomalies diminishes after controlling for multiple testing and improving empirical design. Nevertheless, characteristic-based portfolios continue to exhibit substantial cross-sectional structure, making them informative tools for evaluating asset pricing models.

This thesis employs anomaly portfolios as training and evaluation assets for stochastic discount factor estimation. Anomaly portfolios serve as informative test assets because they aggregate information across firms, reduce idiosyncratic noise, and span a wide range of return patterns associated with different firm characteristics. From an asset pricing perspective, a candidate SDF that prices a large set of anomaly portfolios well is subject to strong no-arbitrage discipline. Accordingly, anomaly portfolios provide a demanding and economically meaningful environment for comparing linear and nonlinear SDF estimates.

2.3 High-Dimensional SDF Estimation and Regularization

2.3.1 Linear factor models and the mean-variance efficient SDF

A key insight into stochastic discount factor estimation in high-dimensional settings is provided by Kozak et al. (2020). The authors show that the mean-variance efficient stochastic discount factor can often be well approximated using linear combinations of asset returns, even when the cross-section of assets is large. From this perspective, traditional linear factor models can be interpreted as approximations to the efficient SDF, with pricing performance determined by how closely the estimated SDF approaches the mean-variance frontier.

2.3.2 Regularization and model selection in asset pricing

Kozak et al. (2020) emphasize that the primary challenge in estimating an efficient SDF is not model flexibility but statistical noise. In large cross-sections, unconstrained estimation leads to severe overfitting and poor out-of-sample performance. Regularization techniques, such as shrinkage and dimension reduction, play a central role in stabilizing SDF estimates and improving pricing accuracy. These methods allow researchers to trade off bias and variance in a controlled manner while preserving economically meaningful risk pricing.

2.3.3 Implications for the benchmark design

The framework in Kozak et al. (2020) motivates the use of regularized linear benchmarks when evaluating more flexible machine learning models. Ordinary least squares provides a low-bias but high-variance estimator, while penalized regressions and principal component methods introduce shrinkage that can substantially improve out-of-sample performance. This perspective suggests that once linear SDFs are properly regularized, the scope for large improvements in average pricing errors from additional model complexity may be limited.

2.4 Machine Learning in Empirical Asset Pricing

2.4.1 Why ML methods are attractive in asset pricing

Machine learning methods provide a natural framework for asset pricing problems characterized by high-dimensional information sets and weak predictive signals. As emphasized by Nagel (2021), traditional linear regression techniques are ill-suited to environments in which the number of potentially relevant predictors is large relative to the available cross-sectional sample size. In such settings, ordinary least squares estimation is prone to severe overfitting, resulting in unstable out-of-sample performance. Machine learning methods address these challenges through regularization, dimension reduction, and flexible function approximation, allowing researchers to incorporate rich information sets without imposing ad hoc sparsity restrictions. Consistent with this perspective, Gu et al. (2020) compare a wide range of machine learning methods to traditional linear benchmarks in forecasting the cross-section of U.S. stock returns. Using a large set of firm characteristics, they document substantial improvements in out-of-sample predictive performance relative to ordinary least squares. These results highlight the ability of machine learning methods to capture nonlinearities and interaction effects that are difficult to model within conventional linear frameworks.

2.4.2 Prediction versus pricing objectives

A central theme in the machine learning asset pricing literature is the distinction between return prediction and asset pricing objectives. As stressed by Nagel (2021), minimizing forecast error for realized returns is not equivalent to identifying economically meaningful risk premia or a valid stochastic discount factor. Asset pricing is concerned with explaining differences in expected returns across assets, which are first-moment objects, whereas predictive accuracy is largely driven by second-moment variation arising from unforeseeable shocks. As a result, machine learning models that perform well in return prediction may yield limited improvements in pricing performance unless their estimation is disciplined by economic structure.

This distinction has motivated subsequent work emphasizing that objectives based on prediction error minimization and no-arbitrage pricing restrictions are fundamentally different, and that gains in return predictability need not translate into proportional reductions in pricing errors across assets.

Against this backdrop, Gu et al. (2020) demonstrate that machine learning methods deliver substantial improvements in return predictability, while also emphasizing the importance of evaluating these gains through an asset pricing lens. Rather than treating predicted returns as an end in themselves, they map forecasts of expected returns into implied stochastic discount factors and assess performance using portfolio-level pricing diagnostics. This mapping provides a bridge between predictive success and asset pricing relevance, but it also implicitly assumes that improved forecasting accuracy translates into improved pricing performance, an assumption that is not guaranteed in general.

2.5 Deep Learning Approaches to Learning the SDF

2.5.1 Learning pricing objects from return forecasts

Rather than estimating the stochastic discount factor directly, Gu et al. (2020) infer pricing implications from machine learning forecasts of expected returns. In this framework, the SDF is constructed as the return on a dynamically managed portfolio that loads on the predicted cross-section of returns. This approach links modern predictive machine learning methods to traditional asset pricing concepts and provides a bridge between forecasting and pricing-based evaluation.

2.5.2 Learning an SDF from large information sets

Recent work has framed SDF estimation as a high-dimensional supervised learning problem Nagel (2021). A prominent implementation of this idea is provided by Chen et al. (2021), who estimate the stochastic discount factor directly as a nonlinear function of firm characteristics

and macroeconomic information using deep neural networks. Their framework addresses key challenges in empirical asset pricing, including the unknown functional form of the pricing kernel, time variation in risk premia, and the low signal-to-noise ratio in individual asset returns. By modeling the SDF itself rather than expected returns, the approach aligns the learning objective directly with the fundamental pricing equation implied by no-arbitrage.

2.5.3 Imposing economic structure or no-arbitrage discipline

Despite their flexibility, machine learning methods require economic discipline to yield meaningful asset pricing implications. As emphasized by Nagel (2021), successful applications of machine learning in asset pricing must incorporate prior economic structure, such as no-arbitrage restrictions or portfolio efficiency considerations, because minimizing prediction error alone risks fitting noise rather than economically relevant variation in expected returns. This insight motivates the use of pricing-based objective functions and moment restrictions when estimating stochastic discount factors with machine learning tools. Consistent with this perspective, Chen et al. (2021) explicitly incorporate no-arbitrage pricing conditions into the learning objective by training neural networks to satisfy conditional moment restrictions rather than forecast realized returns. By disciplining model flexibility with economic constraints, their approach extracts economically meaningful risk premia from noisy return data and demonstrates that off-the-shelf predictive machine learning methods can perform poorly in asset pricing applications unless guided by economic structure.

2.5.4 Evaluation practices in the literature

The machine learning asset pricing literature emphasizes pricing-based evaluation rather than predictive loss as the primary criterion for model assessment. Nagel (2021) stresses that economically meaningful evaluation should focus on pricing errors, Sharpe ratios of stochastic discount factor–implied portfolios, and cross-sectional explanatory power, as these metrics directly reflect the validity of a candidate pricing kernel. Consistent with this view, Chen

et al. (2021) assess model performance using out-of-sample SDF Sharpe ratios, explained average returns, and pricing errors across characteristic-sorted portfolios, explicitly prioritizing no-arbitrage consistency over return forecast accuracy. In contrast, Gu et al. (2020) emphasize portfolio-level performance derived from predicted expected returns, which, while informative about investor value, may mask firm-level pricing inaccuracies, motivating the complementary focus on pricing errors and cross-sectional fit adopted in this thesis.

2.5.5 Where anomaly portfolios fit in this literature

An important contribution of Chen et al. (2021) is their emphasis on identification through the use of anomaly-sorted portfolios as informative test assets. They argue that standard test assets may fail to fully identify the pricing kernel and propose an adversarial procedure that constructs portfolios which are hardest to price, highlighting the role of anomaly portfolios as stress tests for candidate stochastic discount factors. In Gu et al. (2020), anomaly-related firm characteristics play a central role in model construction, serving both as inputs to the machine learning forecasts and as the basis for characteristic-sorted portfolio evaluation, thereby aggregating firm-level information to reduce idiosyncratic noise and sharpen economically meaningful signals. More broadly, the machine learning asset pricing literature frequently relies on anomaly-sorted portfolios as test assets because they provide a structured summary of cross-sectional return variation and help guard against models that appear successful only by fitting a narrow subset of assets. This thesis adopts this perspective by using anomaly portfolios as stress tests to evaluate the pricing implications and identification properties of alternative stochastic discount factors.

2.6 Summary and Positioning of This Thesis

2.6.1 What this thesis does

This thesis studies how the choice of predictive model affects the properties and performance of implied stochastic discount factors in a unified empirical framework. Using a common set of anomaly portfolios as training and evaluation assets, the analysis compares linear and nonlinear models — including ordinary least squares, LASSO, principal component regression, gradient boosting, feedforward neural networks, and recurrent neural networks — under a consistent data construction, training protocol, and evaluation strategy.

Rather than focusing on return forecast accuracy alone, the thesis evaluates models through their implied pricing kernels. Model performance is assessed using out-of-sample pricing errors, cross-sectional explanatory power, and related asset pricing diagnostics. This design ensures that comparisons across models are aligned with the no-arbitrage restrictions that underpin asset pricing theory.

By holding the information set, test assets, and evaluation criteria fixed, the analysis isolates the role of model flexibility in shaping the estimated stochastic discount factor. This approach allows for a clean comparison of how different modeling choices affect both the dynamics of state pricing and overall pricing accuracy.

2.6.2 How it differs from prior work

While recent machine learning studies emphasize the gains from nonlinear modeling, the results in this thesis are consistent with the insights of Kozak et al. (2020). In particular, the analysis shows that once regularization is properly applied, linear and nonlinear models yield stochastic discount factors that are broadly similar in terms of average pricing performance. Differences across models primarily manifest in the dynamics and state dependence of risk pricing rather than in large reductions in firm-level pricing errors.

Chapter 3

Data

3.1 Data Sources and Construction

This chapter describes the datasets used in the empirical analysis, the construction of the return universe, and the preprocessing steps applied prior to model estimation. The primary data sources include equity return and firm-level information obtained from the Wharton Research Data Services (WRDS) platform, macroeconomic data from the Federal Reserve Economic Data (FRED) database, industry classifications derived from Occupational Safety and Health Administration (OSHA) SIC groupings, and monthly anomaly portfolio returns provided by the original database authors.

Monthly stock-level data are obtained from WRDS and include returns, prices, market capitalization, and firm identifiers for U.S. equities. The dataset is structured at the firm-month level and spans a sample period of 2000-01 through 2024-12. Firm identifiers and metadata, including company names and Standard Industrial Classification (SIC) codes, are sourced from WRDS and merged to the return data using consistent firm identifiers.

Monthly Fama–French five-factor data are also obtained from WRDS and used as benchmark factor returns and conditioning variables in the empirical analysis. Macroeconomic data are obtained from the Federal Reserve Economic Data (FRED) database. In particu-

lar, the effective federal funds rate is used as a proxy for the short-term risk-free rate and aligned to the monthly frequency of the equity return data.

Industry classifications are constructed using Standard Industrial Classification (SIC) codes and mapped to broader industry groupings using data from the Occupational Safety and Health Administration (OSHA). This mapping provides a consistent industry taxonomy across firms and time, facilitating aggregation and robustness analyses.

Monthly anomaly portfolio returns are obtained from the anomaly portfolio database constructed by Chen and Zimmermann (2022). These portfolios are formed by sorting stocks on a wide range of firm characteristics and represent systematic return patterns documented in the asset pricing literature. [1]

In this thesis, anomaly portfolios serve as structured test assets and training inputs for learning stochastic discount factors. By aggregating firm-level signals into portfolio returns, these assets provide a rich yet tractable representation of cross-sectional variation in expected returns.

3.2 Sample Construction and Filtering

3.2.1 Sample period and universe

The final empirical sample spans 2000-01 through 2024-12 and includes all historical S&P 500 stocks with available return and firm characteristic data over the sample period.

3.2.2 Rolling train–validation–test windows

To evaluate return predictability and implied stochastic discount factors in a strictly out-of-sample setting, the empirical sample is partitioned into a sequence of rolling train, validation,

[1] Since the anomaly portfolios were initially downloaded for this project in early 2025, the database interface and the underlying portfolio contents have undergone significant revisions. All results in this thesis are based on the original vintages of the anomaly portfolios used at the time of data collection.

and test windows. Each window consists of three non-overlapping subperiods used for model training, hyperparameter selection, and out-of-sample evaluation, respectively.

Specifically, the analysis employs six rolling windows. In each window, models are trained using a fixed-length training sample, validated on a subsequent period, and evaluated on a forward test period. The windows advance forward in time with partial overlap, ensuring that all predictions and estimated pricing kernels are formed using only information available at the time.

This rolling-window structure serves two purposes. First, it prevents look-ahead bias by strictly separating training and evaluation samples. Second, it allows the stability and economic behavior of estimated stochastic discount factors to be examined across multiple market environments. Firms with insufficient data coverage in any of the training, validation, or test subperiods are excluded to maintain consistency across windows.

3.2.3 Data cleaning and alignment

Standard data cleaning procedures are applied, including the removal of observations with missing key variables, alignment of trading dates across datasets, and consistency checks across firm identifiers. Furthermore, firms with an insufficient number of monthly observations in any of the training, validation, or test folds are removed.

3.3 Descriptive Statistics

This section presents summary statistics for the main variables used in the analysis, including equity returns, firm characteristics, factor returns, and anomaly portfolio returns. Summary statistics are reported for both the full sample and relevant subsamples where appropriate.

Table 3.1: Summary Statistics of Predictor Variables

	Count	Mean	Std. Dev.	Min	Max
Panel A: Fama–French Factors					
Market excess return (MKT)	206	0.0072	0.0478	-0.1884	0.1273
SMB	206	-0.0006	0.0276	-0.0963	0.0777
HML	206	-0.0018	0.0345	-0.1573	0.1285
RMW	206	0.0032	0.0194	-0.0489	0.0751
CMA	206	0.0001	0.0195	-0.0692	0.0797
Risk-free rate	206	0.0009	0.0014	0.0000	0.0048
Momentum (UMD)	206	0.0006	0.0469	-0.2997	0.1489
Panel B: Macroeconomic Variable					
Federal funds rate	206	0.2549	0.3443	0.0100	1.1943
Panel C: Anomaly Portfolio Returns (Averaged)					
Anomaly portfolios ($n = 199$)	206	0.0084	0.0483	-0.1942	0.1343

For completeness, the Fama–French factors reported in Panel A are defined as follows. The market excess return (MKT) is the value-weighted return on the market portfolio in excess of the risk-free rate. SMB (Small Minus Big) captures the size premium by taking the return difference between small- and large capitalization firms. HML (High Minus Low) measures the value premium, defined as the return spread between firms with high and low book-to-market ratios. RMW (Robust Minus Weak) captures the profitability premium based on operating profitability, while CMA (Conservative Minus Aggressive) reflects the investment premium associated with firms that invest conservatively versus aggressively. Momentum (UMD) represents the return spread between stocks with high and low prior returns.

Panel C reports summary statistics aggregated across all anomaly-sorted portfolio returns. Rather than listing individual anomaly portfolios, the table reports the average distributional properties across the full set of anomaly signals to convey the scale and dispersion of the high-dimensional predictor set while maintaining tractability.

Chapter 4

Methodology and Empirical Results

4.1 Overview and Empirical Roadmap

This chapter presents the empirical methodology used to estimate stochastic discount factors implied by alternative machine learning models for expected stock returns and reports the resulting empirical findings. Using a common data set, fold structure, and evaluation framework, I compare linear and nonlinear models in terms of their implied pricing kernels, firm-level risk exposures, and cross-sectional pricing performance.

The empirical analysis proceeds in four steps. First, a set of return forecasting models is estimated using rolling out-of-sample folds. Second, the forecasted returns are mapped into stochastic discount factors using no-arbitrage restrictions. Third, the time-series and cross-sectional properties of the estimated stochastic discount factors are examined. Finally, the models are evaluated based on firm-level pricing errors and cross-sectional fit.

4.2 Return Forecasting Models and Out-of-Sample Design

This section describes the return forecasting models considered in the empirical analysis and the out-of-sample design used for their estimation. All models predict next-period excess returns using the same information set and are estimated using identical rolling window folds.

The linear benchmark models include ordinary least squares regression, principal component regression, and LASSO regression. These models impose linear relationships between firm characteristics and expected returns but differ in how they handle high-dimensional predictor sets. Ordinary least squares serves as a baseline specification, while principal component regression and LASSO introduce dimensionality reduction and regularization, respectively.

To allow for nonlinearities and interaction effects among firm characteristics, gradient boosting models are also considered. These models construct ensembles of decision trees and can flexibly adapt to heterogeneous return patterns across firms while remaining purely cross-sectional in nature.

The analysis further includes feedforward neural networks and recurrent neural networks. Feedforward neural networks allow for general nonlinear mappings from firm characteristics to expected returns. Recurrent neural networks extend this framework by incorporating temporal dependencies, enabling expected returns to depend on latent economic states inferred from historical information.

All models are estimated using a rolling window design to avoid look-ahead bias and to reflect the information available to an investor at each point in time. Each fold consists of a training window used for model estimation, followed by an evaluation window over which model predictions are assessed. To ensure comparability, all models use identical folds, predictor sets, and firm universes. Differences in empirical performance can therefore be

attributed solely to differences in model structure rather than differences in data availability or estimation windows.

4.3 From Expected Returns to the Stochastic Discount Factor

Under the absence of arbitrage, asset prices satisfy the conditional moment restriction

$$\mathbb{E}_t \left[M_{t+1} R^e_{t+1,i} \right] = 0, \tag{4.1}$$

where M_{t+1} denotes the stochastic discount factor and $R^e_{t+1,i}$ denotes the excess return of asset i. Forecasted expected returns therefore impose restrictions on the stochastic discount factor through their covariance with asset payoffs.

For each model and fold, the stochastic discount factor is constructed as the mean-variance efficient linear combination of asset returns implied by the model's expected return forecasts. Covariance matrices used in this construction are estimated using training data only to preserve the out-of-sample integrity of the analysis.

4.4 Properties of Estimated Stochastic Discount Factors

This section documents the empirical properties of the stochastic discount factors implied by alternative return forecasting models, focusing on their time-series dynamics and distributional characteristics.

Figure 4.1 plots the time-series of the estimated stochastic discount factors implied by selected linear and nonlinear models. Each series is normalized to have unit mean within each fold to facilitate direct comparison of their dynamics over time.

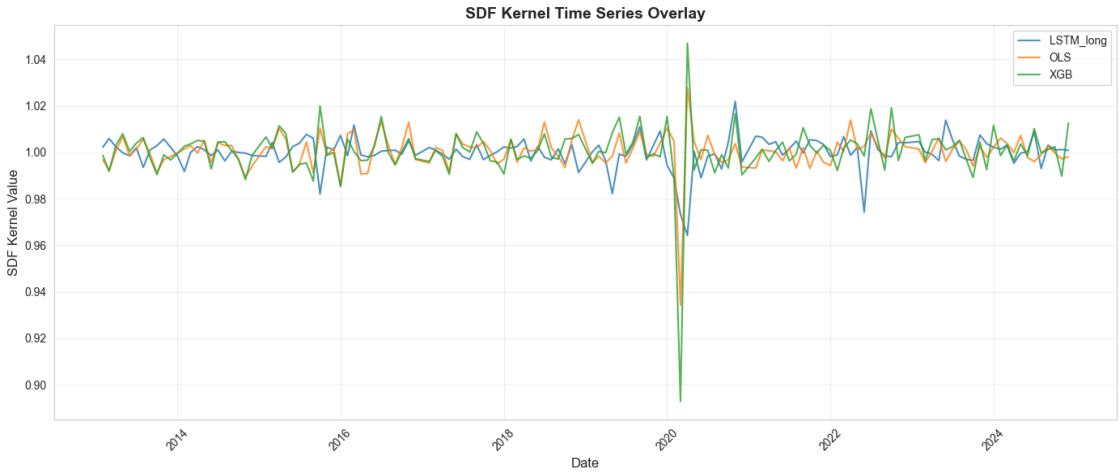

Figure 4.1: Time-series behavior of estimated stochastic discount factors

Across models, the estimated stochastic discount factors display similar low-frequency movements and remain close to one for most of the sample period. However, during episodes of heightened market volatility, differences in dispersion become more pronounced. In particular, nonlinear specifications exhibit larger temporary deviations during periods of market stress, most notably during the COVID-19 crisis.

Table 4.1 reports summary statistics of the estimated stochastic discount factors across models, including measures of dispersion and higher-order moments.

Table 4.1: Summary Statistics of Estimated Stochastic Discount Factors

	Count	Mean	Std. Dev.	Min	25%	50%	75%
LSTM (Long)	138	1.000043	0.006729	0.964259	0.998242	1.000626	1.003173
OLS	138	1.000354	0.008190	0.934256	0.996810	1.000891	1.004038
LASSO	138	1.000614	0.012800	0.893408	0.996859	1.000517	1.004686
PCR	138	1.000268	0.013146	0.895856	0.994990	0.999375	1.004703
XGB	138	1.000534	0.012083	0.892735	0.996729	1.000359	1.005181
FNN	138	1.000359	0.012044	0.902154	0.996463	1.000189	1.004590

	Max	Skewness	Kurtosis
LSTM (Long)	1.021910	-1.871959	8.832631
OLS	1.027676	-3.524748	30.850947
LASSO	1.068474	-3.188256	40.778246
PCR	1.053351	-3.013808	30.174022
XGB	1.046939	-4.730812	47.257118
FNN	1.063238	-2.898679	36.705331

Across all specifications, the estimated stochastic discount factors are tightly centered around one but exhibit substantial asymmetry and excess kurtosis, reflecting infrequent but large deviations from the mean. Differences across models are most pronounced in higher-order moments rather than in average levels.

4.5 Firm-Level Risk Exposures and Pricing Performance

Firm-level risk exposures are measured by the conditional covariance between firm returns and the stochastic discount factor. These exposures quantify the extent to which individual firms perform poorly in states where marginal utility is high and therefore command higher expected returns.

Figure 4.2 presents the cross-sectional distribution of estimated stochastic discount factor betas for each model. Across all models, the distribution of exposures is centered on positive values, indicating that most firms exhibit positive covariance with the pricing kernel. Nonlinear models produce slightly more dispersed exposure distributions, with heavier tails relative to linear benchmarks, suggesting greater heterogeneity in how firms load on adverse economic states.

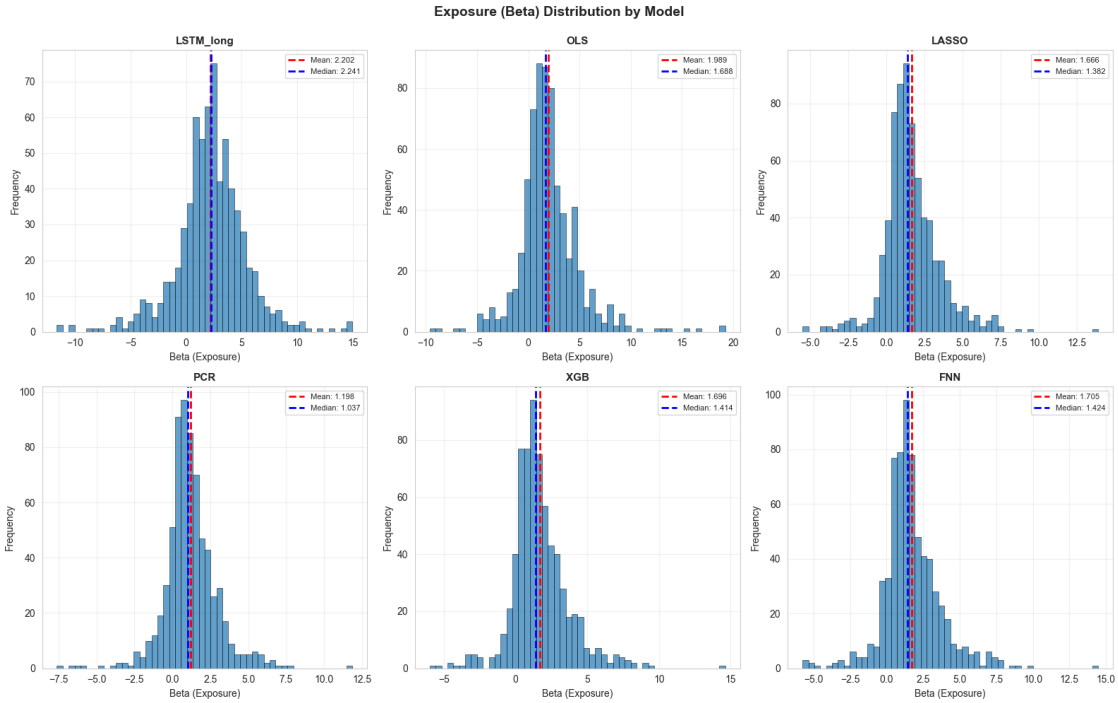

Figure 4.2: Model by model firm exposure distributions with mean and median

Model performance is evaluated using firm-level pricing errors derived from the implied stochastic discount factors. Metrics include mean absolute pricing errors and cross-sectional explanatory power. Table 4.2 reports summary statistics of firm-level absolute pricing errors implied by each stochastic discount factor.

Table 4.2: Summary Statistics of Firm-Level Pricing Errors and Cross-Sectional Fit

Model	Mean($\|\alpha\|$)	Median($\|\alpha\|$)	Std. Dev. ($\|\alpha\|$)	Min($\|\alpha\|$)	Max($\|\alpha\|$)	CS R^2	N_{firms}
LASSO	0.011467	0.009201	0.009959	0.000028	0.088016	0.0966	749
FNN	0.011463	0.009238	0.009968	0.000041	0.088014	0.0925	749
PCR	0.011463	0.009188	0.009989	0.000006	0.088228	0.0533	749
XGB	0.011459	0.009180	0.009966	0.000073	0.088179	0.0946	749
LSTM (Long)	0.011439	0.009145	0.010030	0.000003	0.088766	0.0595	749
OLS	0.011433	0.009178	0.009997	0.000052	0.088257	0.1138	749

Notes: Pricing errors are measured as firm-level alphas implied by each stochastic discount factor. Cross-sectional R^2 measures the proportion of variation in average firm returns explained by the model-implied expected returns.

Figure 4.3 displays the cross-sectional distribution of firm-level pricing errors implied by

each stochastic discount factor.

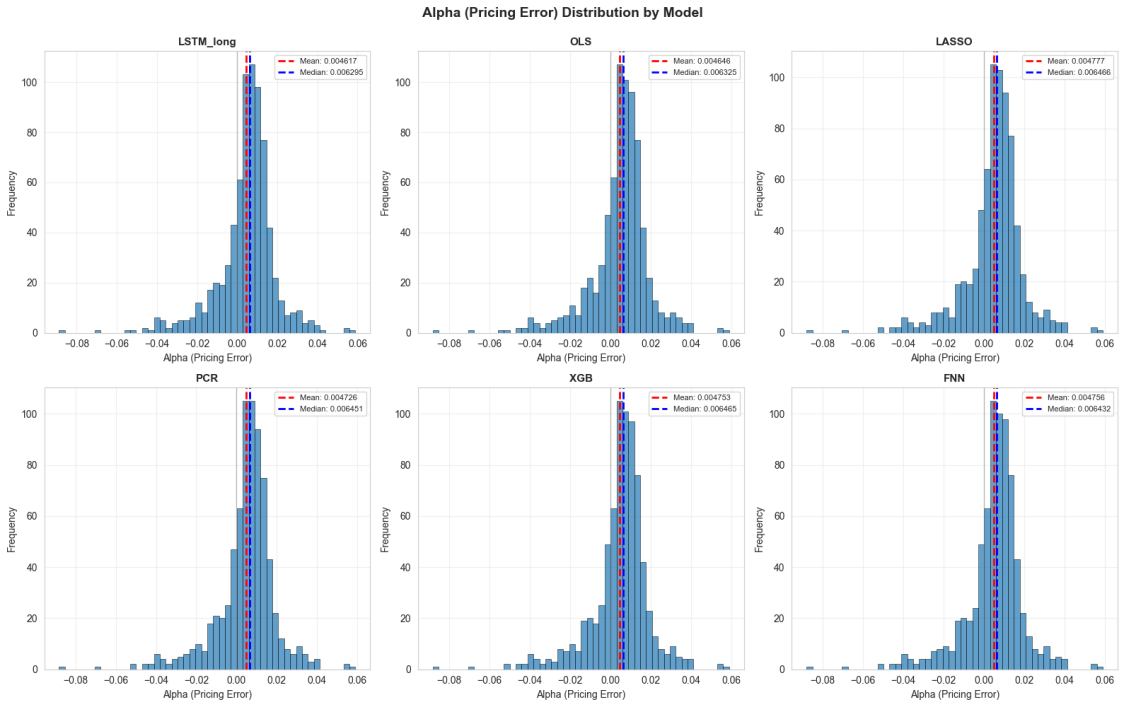

Figure 4.3: Model-by-model firm pricing error distributions with mean and median

Across models, pricing errors are tightly centered around zero and exhibit similar dispersion and tail behavior. Mean and median pricing errors remain closely aligned across specifications, indicating that increased model flexibility does not translate into materially improved firm-level pricing accuracy.

Chapter 5

Discussion

5.1 Purpose and Overview

As a reminder, the purpose of this thesis was to examine whether different machine learning models, when used to forecast firm-level equity returns, imply materially different stochastic discount factors, and whether such differences translate into economically meaningful variation in firm-level risk exposures and pricing errors. The empirical results indicate that most firms exhibit positive covariance with the estimated pricing kernel and that absolute pricing errors are highly similar across model classes. While nonlinear models imply more state-dependent and time-varying stochastic discount factors, these differences do not translate into substantial reductions in firm-level pricing errors relative to linear benchmarks. This chapter interprets these findings through the lens of asset pricing theory, focusing on economic meaning, identification, and limitations, rather than introducing new estimation or empirical results.

5.2 Summary of Empirical Findings

5.2.1 Properties of the Estimated Stochastic Discount Factors

Across all model specifications, the estimated stochastic discount factors exhibit similar unconditional means and remain economically admissible, with no evidence of implausible magnitudes or instability. This comparability ensures that subsequent differences in higher-order moments reflect differences in implied state pricing rather than mechanical scale effects. With respect to how aggressively each model reweights economic states, nonlinear models generate stochastic discount factors with greater dispersion and heavier tails, as reflected in higher unconditional volatility and kurtosis. Through this increased volatility and tail thickness, nonlinear models place relatively greater weight on adverse economic states in the pricing kernel. However, these differences primarily affect the dynamics and state dependence of the stochastic discount factor, rather than its overall level, a distinction that helps explain why greater flexibility does not translate into large differences in average firm-level pricing errors.

5.2.2 Firm-Level Risk Exposures

Across models, firm-level exposures to the stochastic discount factor are predominantly positive, indicating that the average firm tends to deliver lower returns in states where the pricing kernel is high. This pattern is consistent with standard asset pricing intuition, in which firms that perform poorly in adverse economic states command positive risk premia. Noting earlier where nonlinear specifications imply greater cross-sectional dispersion in SDF betas, model flexibility primarily affects the identification of of firms with extreme exposure to aggregate risk, not the average level of exposure across the cross-section.

Economically, higher exposure to the stochastic discount factor reflects greater sensitivity to states in which the marginal utility of consumption is high. Firms with larger SDF betas tend to underperform precisely when consumption risk is mostly costly, making them

undesirable hedges and requiring compensation in equilibrium

5.2.3 Pricing Errors and Cross-Sectional Fit

This pattern reflects identification limits inherent in firm-level asset pricing tests. Average pricing errors are unconditional, cross-sectional objects and therefore depend primarily on unconditional covariances between returns and the stochastic discount factor. As a result, models that differ substantially in how they price extreme economic states may nevertheless deliver similar average pricing errors when evaluated over the full sample. Flexible machine learning methods primarily affect the dynamics and tail behavior of the pricing kernel, refining how risk is priced in adverse states rather than improving unconditional cross-sectional fit. Consequently, statistically detectable differences in cross-sectional R^2 need not correspond to economically meaningful reductions in firm-level mispricing, underscoring the distinction between statistical alignment and economic explanation.

5.3 Economic Interpretation

5.3.1 Economic Meaning of the Stochastic Discount Factor

From an economic perspective, the stochastic discount factor summarizes how the market prices risk across states of the world by assigning higher weight to outcomes in which marginal utility is high. In this framework, adverse economic states receive greater weight because payoffs realized in those states are more valuable to investors. The estimated pricing kernels in this thesis are consistent with this interpretation: firms with higher exposure to the stochastic discount factor tend to underperform in periods of economic stress and therefore command compensation in equilibrium. Differences across models in the shape and dynamics of the pricing kernel reflect alternative ways of mapping firm-level information into state prices, rather than fundamentally different notions of risk. As such, the stochastic discount factor serves as a unifying object that links return predictability, firm-level exposures, and

pricing performance within a coherent asset pricing framework.

5.3.2 Why Model Complexity Affects SDF Dynamics More Than Pricing Accuracy

The finding that nonlinear models imply more state-dependent stochastic discount factors without materially improving average pricing accuracy reflects a fundamental distinction between modeling state prices and improving unconditional fit. Flexible machine learning methods are well suited to capturing nonlinearities and interactions that shape how risk is priced across different economic states. As shown in the empirical analysis, these differences manifest primarily in the tails of the pricing kernel distribution and during periods of elevated market stress, when state prices vary most sharply.

However, average pricing errors depend primarily on unconditional covariances between returns and the pricing kernel, which are already well approximated by simpler linear structures. Because extreme states occur infrequently, improvements in modeling state-dependent pricing dynamics need not translate into large reductions in average pricing errors. In this sense, the empirical results do not suggest that nonlinear models fail to capture relevant pricing information, but rather that their advantages operate through the pricing of rare but economically important states rather than through improvements in unconditional cross-sectional fit.

This distinction highlights an identification tradeoff inherent in empirical asset pricing. Pricing errors aggregate information across all states and therefore place limited weight on extreme outcomes, while the stochastic discount factor itself is identified most strongly by how it behaves in adverse states. By altering the dynamics and tail behavior of the pricing kernel without substantially changing average pricing errors, nonlinear models primarily refine the economic interpretation of risk pricing rather than the statistical fit of the model. This perspective helps reconcile the modest gains in cross-sectional pricing accuracy with the substantial differences observed in the implied pricing of economic states.

5.4 Relation to the Literature

5.4.1 Machine Learning, Return Prediction, and No-Arbitrage Pricing

A large literature documents the ability of machine learning methods to improve the statistical accuracy of return forecasts relative to linear benchmarks. The results of this thesis highlight that such forecasting gains do not necessarily translate into materially improved asset pricing performance, as prices are governed by covariance with the stochastic discount factor rather than predictive accuracy alone. While flexible models imply richer, more state-dependent pricing kernels, average pricing errors remain largely determined by unconditional moment conditions. This distinction reinforces the importance of economic structure and no-arbitrage restrictions when interpreting machine learning approaches in asset pricing.

5.4.2 Linear Benchmarks and Factor Models

The continued competitiveness of linear benchmarks in terms of average pricing performance aligns with a broad body of evidence from factor pricing models. Despite their simplicity, linear structures appear sufficient to capture the primary unconditional covariances that govern firm-level pricing errors. The results here suggest that the main contribution of more flexible models lies in refining the dynamics of state pricing rather than overturning the explanatory power of linear factor representations. Hence, machine learning methods are best understood as extensions of, rather than replacements for, traditional factor-based asset pricing frameworks.

5.5 Robustness, Limitations, and Identification

5.5.1 Sample Construction and Asset Coverage

The empirical analysis focuses on firms in the S&P 500 universe, which provides a large, liquid, and well-covered cross-section of U.S. equities. This choice mitigates concerns related to illiquidity, sparse trading, and data availability, but also limits the scope of the results. In particular, pricing errors and risk exposures may behave differently for smaller or less liquid firms, where information frictions and limits to arbitrage are more pronounced. As a result, the findings should be interpreted as describing the pricing implications of machine learning models within a relatively liquid and well-integrated segment of the equity market.

5.5.2 Model Dependence and Regularization

The results depend on choices related to model architecture, regularization, and sample partitioning, all of which are necessary in high-dimensional settings. Regularization and fold-based estimation play a central role in stabilizing both return forecasts and the implied stochastic discount factors, limiting overfitting while preserving economically meaningful variation. While alternative tuning choices may affect the precise dynamics of the estimated pricing kernels, the qualitative findings regarding firm-level pricing errors and cross-sectional fit are robust across model classes. This suggests that the main conclusions are driven by structural features of the asset pricing problem rather than by specific modeling choices.

5.5.3 Identification Limits

The identification of the stochastic discount factor in this thesis is inherently tied to the set of test assets and moment conditions used in evaluation. Because pricing performance is assessed using unconditional firm-level returns, the empirical results primarily identify average covariances between returns and the pricing kernel rather than the full state-contingent structure of risk prices. As a result, distinct stochastic discount factors that differ in their

dynamics or tail behavior may nonetheless generate similar pricing errors when evaluated on the same asset universe. Accordingly, the results are most informative for comparing how alternative model specifications translate return forecasts into implied firm-level risk exposures and state pricing, rather than for identifying a unique or fully structural stochastic discount factor from asset prices alone.

5.6 Implications for Asset Pricing and Machine Learning

5.6.1 Implications for Empirical Asset Pricing and Machine Learning

The results reinforce a central insight of empirical asset pricing: while flexible models can refine the dynamics of state pricing, average cross-sectional pricing performance remains largely governed by unconditional risk covariances captured by simpler structures. For machine learning applications in finance, this highlights both the value and the limits of flexibility in low signal-to-noise environments, underscoring the importance of economic constraints when translating predictive gains into asset pricing implications.

5.7 Directions for Future Research

Several extensions could further clarify the asset pricing implications of machine learning–based stochastic discount factor estimation. First, expanding the set of test assets to include characteristic-sorted portfolios, less liquid equities, or alternative asset classes may improve identification of state-dependent pricing differences that are muted in a large-cap equity universe. Second, incorporating richer macroeconomic conditioning or alternative estimation frameworks, such as MAXSER-style approaches that emphasize pricing in high-

volatility states, could help sharpen the role of extreme economic conditions in shaping the pricing kernel. Finally, explicitly accounting for transaction costs and estimation uncertainty may further bridge the gap between predictive modeling and economically implementable asset pricing.

5.8 Conclusion

While machine learning models imply richer and more dynamic stochastic discount factors, their pricing implications remain largely governed by familiar economic constraints and identification limits. Taken together, the results indicate that machine learning primarily refines the characterization of state-dependent risk, while leaving average cross-sectional pricing performance largely unchanged.

Supporting Materials

This section provides supplementary material intended to clarify the modeling framework and economic interpretation used throughout the thesis. The goal is to centralize notation, assumptions, and intuition in a single location to support readers seeking additional technical or conceptual context without interrupting the main empirical narrative.

A Model Notation and Assumptions

A.1 Linear Prediction Models

The linear models considered in this thesis include ordinary least squares (OLS), principal component regression (PCR), and LASSO. All models share a common prediction framework in which next-period excess returns are modeled as a linear function of observable predictors,

$$r_{i,t+1} = X'_{i,t}\beta + \varepsilon_{i,t+1},$$

where $X_{i,t}$ denotes a vector of firm-level characteristics observed at time t, β is a vector of coefficients, and $\varepsilon_{i,t+1}$ is an unpredictable innovation.

OLS estimates β by minimizing squared prediction errors without regularization, implicitly assuming a low-dimensional and well-conditioned predictor set. PCR reduces dimensionality by projecting predictors onto a small number of principal components before estimation, imposing a factor structure on expected returns. LASSO introduces an ℓ_1 penalty on the

coefficients, encouraging sparsity and shrinking less informative predictors toward zero.

Across these models, the key maintained assumptions are linearity in predictors, a static mapping from characteristics to expected returns at each point in time, and the absence of explicit state dependence or nonlinear interactions. Regularization in PCR and LASSO is used to improve statistical stability rather than to impose economic structure.

A.2 Nonlinear Prediction Models

The nonlinear models considered in this thesis include gradient boosted trees, feedforward neural networks, and recurrent neural networks with long short-term memory (LSTM). These models relax the linearity assumption by allowing expected returns to be represented as a flexible nonlinear function of predictors,

$$r_{i,t+1} = f(X_{i,t}) + \varepsilon_{i,t+1},$$

where $f(\cdot)$ is an unknown function learned from the data.

Tree-based methods capture nonlinearities and interactions through recursive partitioning of the predictor space. Feedforward neural networks approximate general nonlinear functions using layered transformations of the input features. LSTM models extend this framework by allowing expected returns to depend on latent states that evolve dynamically over time, enabling the model to incorporate information from past observations.

These models trade interpretability for flexibility and rely on architectural constraints and regularization to control overfitting. Unlike linear models, they do not impose an explicit factor structure and allow risk exposures to vary nonlinearly across firms and states of the economy.

B The Stochastic Discount Factor

B.1 What Is the Stochastic Discount Factor?

At a high level, a stochastic discount factor (SDF) describes how investors value a dollar received in different future states of the world. Rather than applying a single discount rate to all future payoffs, the SDF assigns state-dependent weights that reflect the timing and economic conditions under which payoffs are realized.

An intuitive interpretation of the SDF is as a system of implicit state prices. Payoffs delivered in adverse states of the world, such as recessions or periods of financial stress, are typically valued more highly than payoffs delivered in favorable states. As a result, assets that perform well in bad states provide insurance and therefore command lower expected returns, while assets that perform poorly in bad states must offer higher average returns as compensation.

Formally, the SDF m_{t+1} is a random variable that reweights future payoffs so that all traded assets satisfy a common pricing condition,

$$\mathbb{E}[m_{t+1}R_{i,t+1}] = 1.$$

This condition is best understood as a no-arbitrage restriction: current prices reflect a weighted average of future payoffs, where the weights encode investors' preferences over economic states.

B.2 How to Interpret and Contextualize the Stochastic Discount Factor

When working with an estimated stochastic discount factor, it is useful to focus on interpretation rather than mechanics. An estimated SDF should not be viewed as a trading strategy or as a return forecasting device. Instead, it is best understood as a diagnostic object that

reveals which sources of risk are rewarded in the data.

Assets earn risk premia because their returns covary with the SDF. Assets that tend to pay off when the SDF is high, corresponding to adverse economic states, provide insurance and therefore command lower expected returns. Conversely, assets that perform poorly when the SDF is high are exposed to priced risk and must offer higher average returns.

From this perspective, the SDF translates observed covariance patterns into economic statements about risk compensation. If an estimated SDF prices a wide range of assets with small pricing errors, it implies that the dominant dimensions of systematic risk have been successfully identified. Differences across models in this thesis are therefore interpreted as differences in how economic states are priced, rather than as evidence of purely statistical dominance.

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